Section 2.7 Coordinates, vectors & transformations

坐标、向量与变换

Method 1 / 方法 1 [GE07-M01]

Coordinates succeed when they encode the strongest geometry with the fewest variables. Use affine coordinates for incidence and ratios, orthonormal coordinates for lengths and angles, and signed normal equations for sides and containment. A final coordinate solution must reproduce every segment and orientation restriction.

坐标法应以最少变量编码最强几何条件。关联与比例适用仿射坐标，长度与角度适用标准正交坐标，两侧及容纳适用有向法向方程。最后须用坐标还原全部线段及方向限制。

Parameterize lines, midpoints and affine side divisions

参数化直线、中点及仿射分点

Procedure

1. Normalize a convenient triangle, rectangle or parallelogram, preserving only the metric information actually needed.

2. Write a point on a line as $P + tv$ and a side point as $(1-t)A + tB$ with $0 \le t \le 1$.

3. Midpoint and centroid conditions are linear vector averages.

4. Solve the linear incidence equations before any squared-distance equation.

通用步骤

1. 规范方便的三角形、矩形或平行四边形，仅保留实际需要的度量信息。

2. 直线点写为 $P + tv$，边上点写为 $(1-t)A + tB$ 并保留 $0 \le t \le 1$。

3. 中点与重心条件是向量线性平均。

4. 先解线性关联，再使用距离平方方程。

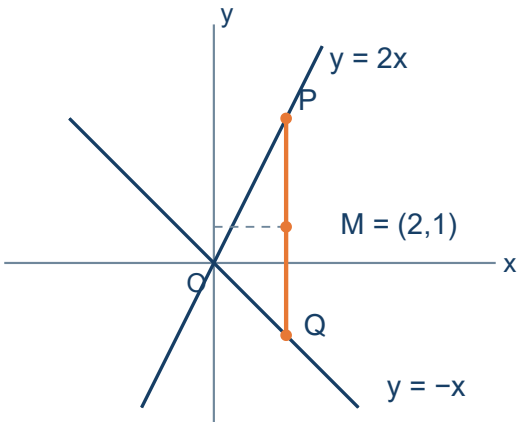
Section 2.7 Coordinates, vectors & transformations

坐标、向量与变换

Method 1 / 方法 1 [GE07-M01] (continued / 续)

> Example. Point P lies on $y = 2x$, point Q on $y = -x$, and their midpoint is $(2, 1)$. Find PQ.

> 例题：P 在 $y = 2x$ 上，Q 在 $y = -x$ 上，中点为 $(2, 1)$，求 PQ。



Worked solution

1. Write $P = (t, 2t), Q = (u, -u)$. Midpoint equations are $t + u = 4, 2t - u = 2$.

2. They give $t = u = 2$, hence $P = (2, 4), Q = (2, -2)$.

3. The segment is vertical of length six; its midpoint is indeed $(2, 1)$.

例题解析

1. 设 $P = (t, 2t), Q = (u, -u)$，中点方程为 $t + u = 4, 2t - u = 2$。

2. 解得 $t = u = 2$，所以 $P = (2, 4), Q = (2, -2)$。

3. 线段竖直，长为六，中点也确为 $(2, 1)$。

Section 2.7 Coordinates, vectors & transformations

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Method 1 / 方法 1 [GE07-M01] (continued / 续)

An affine normalization preserves ratios and area ratios, but not arbitrary lengths or angles.

仿射规范保持比例与面积比，不保持一般长度或角度。

With barycentric coordinates $\alpha+\beta+\gamma=1$, a line parallel to the side opposite A fixes α. Three fixed coordinate levels produce a signed similar triangle with scale $1-a-b-c$.

Align an angle bisector with the x-axis to place adjacent vertices at $(c\cos t, c\sin t), (a\cos t, -a\sin t)$; this keeps the bisector intercept and perpendicular cuts linear.

在重心坐标 $\alpha+\beta+\gamma=1$ 下，平行于 A 对边的直线固定 α。三条固定水平线形成有向比例为 $1-a-b-c$ 的相似三角形。

以角平分线为横轴，可把两邻顶点写为 $(c\cos t, c\sin t), (a\cos t, -a\sin t)$，使平分线截点及垂直截线保持线性。

Practice / 配套练习: AIME-2001-II-Q04 (WB p. 350), AIME-2003-I-Q15 (WB p. 387), AIME-2003-II-Q11 (WB p. 388), AIME-2005-I-Q10 (WB p. 368), AIME-2006-II-Q06 (WB p. 370), AIME-2009-I-Q04 (WB p. 351), AIME-2011-I-Q02 (WB p. 353), AIME-2011-II-Q02 (WB p. 355), AIME-2011-II-Q04 (WB p. 311), AIME-2013-II-Q13 (WB p. 399), AIME-2014-II-Q11 (WB p. 401), AIME-2017-II-Q10 (WB p. 377), AIME-2019-II-Q07 (WB p. 229), AIME-2020-II-Q12 (WB p. 409), AIME-2022-I-Q03 (WB p. 312)

Section 2.7 Coordinates, vectors & transformations

坐标、向量与变换

Method 2 / 方法 2 [GE07-M02]

Equal distances linearize centers and radical axes

等距线性化圆心与根轴

Procedure

1. Expand squared distances to two fixed points and subtract; the quadratic point terms cancel.

2. Find circumcenters from two perpendicular-bisector equations, or circle centers from symmetry and tangency normals.

3. Subtract two circle equations to obtain their radical axis before solving a remaining quadratic.

4. Check positive radius, the distinct second intersection and the required side or ray.

通用步骤

1. 展开到两定点的距离平方并相减，点的二次项会消去。

2. 由两垂直平分线求外心，或由对称及相切法向定位圆心。

3. 两圆方程先相减求根轴，再解剩余二次式。

4. 检查正半径、不同第二交点及所需边或射线。

> Example. Find the circle through $(0,0),(6,0),(0,8)$.

> 例题：求经过 $(0,0),(6,0),(0,8)$ 的圆。

Section 2.7 Coordinates, vectors & transformations

坐标、向量与变换

Method 2 / 方法 2 [GE07-M02] (continued / 续)

Worked solution

1. Let its center be (u, v). Equality of the first two squared distances gives $12u = 36$, so $u = 3$.

2. Equality with the third gives $16v = 64$, so $v = 4$.

3. The radius is $\sqrt{3^2 + 4^2} = 5$, hence $(x-3)^2 + (y-4)^2 = 25$.

例题解析

1. 设圆心 (u, v)，前两点等距给出 $12u = 36$，故 $u = 3$。

2. 与第三点等距给出 $16v = 64$，故 $v = 4$。

3. 半径为五，所以圆方程为 $(x-3)^2 + (y-4)^2 = 25$。

A line of equal powers differs from a perpendicular bisector when the radii differ.

半径不等时，等点幂直线不同于垂直平分线。

Practice / 配套练习: AIME-2001-II-Q06 (WB p. 283), AIME-2001-II-Q07 (WB p. 313), AIME-2002-I-Q15 (WB p. 452), AIME-2005-II-Q08 (WB p. 332), AIME-2006-II-Q09 (WB p. 333), AIME-2008-II-Q11 (WB p. 344), AIME-2009-II-Q05 (WB p. 330), AIME-2009-II-Q10 (WB p. 371), AIME-2011-II-Q13 (WB p. 395), AIME-2012-II-Q15 (WB p. 297), AIME-2014-I-Q15 (WB p. 298), AIME-2014-II-Q08 (WB p. 335), AIME-2017-I-Q10 (WB p. 80), AIME-2019-I-Q13 (WB p. 302), AIME-2019-I-Q15 (WB p. 303), AIME-2019-II-Q11 (WB p. 406), AIME-2020-II-Q15 (WB p. 305), AIME-2022-I-Q08 (WB p. 336), AIME-2023-II-Q09 (WB p. 288), AIME-2024-I-Q05 (WB p. 363), AIME-2025-I-Q08 (WB p. 382), AIME-2025-II-Q01 (WB p. 364), AIME-2025-II-Q06 (WB p. 339)

Section 2.7 Coordinates, vectors & transformations

坐标、向量与变换

Method 3 / 方法 3 [GE07-M03]

Dot and cross products encode metric constraints

点积与叉积编码度量约束

Procedure

1. Use $u \cdot v = |u||v| \cos\theta$ for angles and perpendicularity.

2. Use the planar determinant or spatial cross product for signed area and orientation.

3. Divide cross by dot only after checking the dot sign and whether the requested angle is acute.

4. Expand squared lengths to expose cancellations, closure identities or Gram determinants.

通用步骤

1. 用 $u \cdot v = |u||v| \cos\theta$ 处理夹角及垂直。

2. 用平面行列式或空间叉积处理有向面积与方向。

3. 叉积除点积之前，检查点积符号及所求是否锐角。

4. 展开距离平方，以显露消项、闭合恒等式或格拉姆行列式。

> Example. For $u = (1, 2), v = (3, 1)$, find their acute angle and the parallelogram area.

> 例题：给定 $u = (1, 2), v = (3, 1)$，求两向量锐夹角及平行四边形面积。

Worked solution

1. The dot product is five and the determinant is $1 - 6 = -5$. Both vectors are nonzero and the dot is positive.

Section 2.7 Coordinates, vectors & transformations

坐标、向量与变换

Method 3 / 方法 3 [GE07-M03] (continued / 续)

2. Thus $\tan\theta = |-5|/5 = 1$, so $\theta = 45^\circ$.

3. The parallelogram area is the determinant magnitude five.

例题解析

1. 点积为五，行列式为 $1-6=-5$；两向量非零且点积正。

2. 故 $\tan\theta = 1$，锐角为 45°。

3. 平行四边形面积为行列式绝对值五。

An absolute cross-to-dot ratio does not distinguish an acute angle from its supplement; retain the dot sign.

叉积与点积的绝对值比不能区分锐角及其补角，应保留点积符号。

A closed polygon has side-vector sum zero; squaring this equation relates pairwise side products weighted by direction cosines. In an equiangular pentagon only the72- and144-degree cosine classes remain.

For vectors v_i of equal positive length R and sum zero, $|v_i - x| \geq R - v_i \cdot x/R$ sums to $\sum |v_i - x| \geq nR$, attained at $x = 0$.

For generating vectors u, v, w, the Gram determinant $\det(u_i \cdot u_j)$ equals squared parallelepiped volume. Face diagonals determine pairwise norms and dot products through $|u \pm v|^2$.

闭合多边形的边向量和为零，将其平方可联系按方向余弦加权的两两边积。等角五边形只剩七十二度与一百四十四度两类余弦。

Section 2.7 Coordinates, vectors & transformations

坐标、向量与变换

Method 3 / 方法 3 [GE07-M03] (continued / 续)

若 v_i 均长 $R>0$ 且向量和为零，则 $|v_i-x|\geq R-v_i\cdot x/R$ 相加得到 $\sum|v_i-x|\geq nR$，在 $x=0$ 取等。

生成向量 u,v,w 的格拉姆行列式 $\det(u_i\cdot u_j)$ 等于平行六面体体积平方。面对角线可由 $|u\pm v|^2$ 求范数及点积。

Practice / 配套练习: AIME-2000-II-Q08 (WB p. 366), AIME-2002-II-Q02 (WB p. 436), AIME-2003-II-Q14 (WB p. 389), AIME-2005-II-Q12 (WB p. 392), AIME-2009-II-Q03 (WB p. 352), AIME-2011-II-Q13 (WB p. 395), AIME-2012-I-Q14 (WB p. 92), AIME-2013-I-Q07 (WB p. 373), AIME-2016-I-Q09 (WB p. 183), AIME-2017-II-Q15 (WB p. 405), AIME-2021-I-Q06 (WB p. 449), AIME-2021-II-Q12 (WB p. 410), AIME-2022-I-Q05 (WB p. 20), AIME-2023-I-Q05 (WB p. 362), AIME-2023-I-Q12 (WB p. 411), AIME-2023-I-Q13 (WB p. 460), AIME-2023-II-Q12 (WB p. 412), AIME-2026-I-Q14 (WB p. 414)

Section 2.7 Coordinates, vectors & transformations

坐标、向量与变换

Method 4 / 方法 4 [GE07-M04]

Unit normals give signed distances and projections

单位法向量给出有向距离与投影

Procedure

1. For line $ax+by=c$, use signed distance $(ax+by-c)/\sqrt{a^2+b^2}$.

2. The perpendicular foot is $P-[(n\cdot P-c)/|n|^2]n$.

3. Use the sign to identify a half-plane and retain the normalization when the line direction varies.

4. The spatial plane formula is identical with a three-component normal.

通用步骤

1. 直线 $ax+by=c$ 的有向距离为 $(ax+by-c)/\sqrt{a^2+b^2}$。

2. 垂足为 $P-[(n\cdot P-c)/|n|^2]n$。

3. 利用正负识别半平面，直线方向变化时仍须保留归一化。

4. 空间平面公式相同，只需使用三分量法向量。

> Example. Find the distance and perpendicular foot from $(4,3)$ to $3x+4y=10$.

> 例题：求点 $(4,3)$ 到直线 $3x+4y=10$ 的距离及垂足。

Worked solution

1. The normal is $(3,4)$ of length five, and the residual is $12+12-10=14$. Thus distance is $14/5$.

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坐标、向量与变换

Method 4 / 方法 4 [GE07-M04] (continued / 续)

2. The foot is $(4,3)-(14/25)(3,4)=(58/25,19/25)$.

3. Substitution gives $3(58/25)+4(19/25)=10$, checking incidence.

例题解析

1. 法向量为 $(3,4)$、长五，残量为十四，故距离为 $14/5$。

2. 垂足为 $(4,3)-(14/25)(3,4)=(58/25,19/25)$。

3. 代入 $3x+4y=10$ 成立，核实垂足在线上。

Distance to a segment needs an endpoint check if the foot leaves the segment.

垂足若离开线段，求线段距离还须比较端点。

Practice / 配套练习: AIME-2001-I-Q12 (WB p. 431), AIME-2004-I-Q10 (WB p. 769), AIME-2005-I-Q14 (WB p. 391), AIME-2006-I-Q14 (WB p. 393), AIME-2008-I-Q14 (WB p. 186), AIME-2011-I-Q03 (WB p. 354), AIME-2013-I-Q12 (WB p. 398), AIME-2015-II-Q11 (WB p. 402), AIME-2015-II-Q15 (WB p. 345), AIME-2016-I-Q14 (WB p. 403), AIME-2018-I-Q08 (WB p. 378), AIME-2019-II-Q13 (WB p. 407), AIME-2021-I-Q09 (WB p. 380), AIME-2023-I-Q08 (WB p. 381), AIME-2026-I-Q12 (WB p. 434), AIME-2026-II-Q12 (WB p. 435)

Section 2.7 Coordinates, vectors & transformations

坐标、向量与变换

Method 5 / 方法 5 [GE07-M05]

Rotations and reflections preserve lengths but change orientation

旋转与反射保持长度但改变方向

Procedure

1. Use $R_\theta(x,y)=(x\cos\theta-y\sin\theta, x\sin\theta+y\cos\theta)$.

2. A rotation about C maps P to $C+R_\theta(P-C)$; compare edge vectors to recover its angle first.

3. For squares use a signed quarter-turn, and for equilateral apices use the two signed60-degree turns.

4. A fold crease is the perpendicular bisector of a point and its image; repeated folds also reverse layer order.

通用步骤

1. 使用 $R_\theta(x,y)=(x\cos\theta-y\sin\theta, x\sin\theta+y\cos\theta)$。

2. 绕 C 旋转为 $C+R_\theta(P-C)$，先比较边向量确定角度。

3. 正方形用带符号四分之一转，等边顶点用正负六十度转。

4. 折痕是点与其像的垂直平分线，重复折叠还会颠倒层序。

> Example. A 90° counterclockwise rotation sends $(0,0)$ to $(3,1)$. Find its center and the image of $(1,0)$.

> 例题：一次逆时针九十度旋转将 $(0,0)$ 映为 $(3,1)$，求旋转中心及 $(1,0)$ 的像。

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Method 5 / 方法 5 [GE07-M05] (continued / 续)

Worked solution

1. For center (u, v), the origin image is $(u+v, v-u) = (3, 1)$.

2. Solve to get $(u, v) = (1, 2)$.

3. The image of $(1, 0)$ is $(1, 2) + R_{90}(0, -2) = (3, 2)$.

例题解析

1. 若中心为 (u, v)，原点像为 $(u+v, v-u) = (3, 1)$。

2. 解得中心 $(1, 2)$。

3. 点 $(1, 0)$ 的像为 $(1, 2) + R_{90}(0, -2) = (3, 2)$。

Both signed rotations often solve length equations; use the specified quadrant or side of a line.

正负旋转常都满足长度方程，应按象限或直线两侧筛选。

Folding the right half of an L-cell strip of thickness H over the left maps a right-half cell position p to $L+1-p$ and bottom-up layer b to $2H+1-b$; left-half data remain unchanged before updating L, H.

For a rolling marker, compute the moving center's orbital position and then add the marker's ground-frame spin offset. A full spin restores the marker direction, not the center position.

将长 L 格、厚 H 层纸条右半折到左半上方时，右半格位置 p 变为 $L+1-p$，自底层号 b 变为 $2H+1-b$；左半数据不变，然后更新 L, H。

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Method 5 / 方法 5 [GE07-M05] (continued / 续)

滚动标记点先算圆心公转位置，再加地面参照下的自转偏移。自转整周恢复的是标记方向，不是圆心位置。

Practice / 配套练习: AIME-2000-I-Q02 (WB p. 349), AIME-2004-II-Q07 (WB p. 367), AIME-2004-II-Q15 (WB p. 390), AIME-2011-I-Q14 (WB p. 394), AIME-2012-I-Q13 (WB p. 396), AIME-2012-II-Q13 (WB p. 397), AIME-2013-I-Q09 (WB p. 271), AIME-2013-II-Q04 (WB p. 356), AIME-2014-I-Q10 (WB p. 374), AIME-2015-I-Q04 (WB p. 357), AIME-2015-I-Q07 (WB p. 375), AIME-2019-II-Q01 (WB p. 359), AIME-2020-II-Q04 (WB p. 360), AIME-2022-II-Q11 (WB p. 327), AIME-2025-I-Q02 (WB p. 245), AIME-2025-I-Q09 (WB p. 417), AIME-2025-I-Q14 (WB p. 190), AIME-2026-I-Q05 (WB p. 66), AIME-2026-I-Q10 (WB p. 383)

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Method 6 / 方法 6 [GE07-M06]

Centroid averages expose a common homothety

重心平均显露共同位似

Procedure

1. Write midpoint and centroid vectors as exact averages.

2. Rearrange all target points into $T_i = O + t(V_i - O)$ with one center and scale.

3. Use scale $|t|$ for length, t^2 for planar area and $|t|^3$ for volume.

4. For a centrally symmetric family, pair opposite points to prove equal area or a center bound.

通用步骤

1. 把中点及重心向量写为精确平均。

2. 将各目标点整理为同一中心、同一比例的 $T_i = O + t(V_i - O)$。

3. 长度用 $|t|$，面积用 t^2，体积用 $|t|^3$。

4. 中心对称点族可配对相反点，证明等面积或圆心界。

> Example. Let P be inside a unit square. The centroids of its four triangles with the square sides form a quadrilateral. Find its area.

> 例题：P 为单位正方形内部点，P 与四条边分别组成三角形，其四重心组成四边形，求面积。

Worked solution

1. If M_i is a side midpoint, its fan-triangle centroid is $T_i = (P + 2M_i)/3$.

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Method 6 / 方法 6 [GE07-M06] (continued / 续)

2. Thus the centroid quadrilateral is a homothety of the midpoint quadrilateral with linear scale $2/3$.

3. The midpoint quadrilateral has area $1/2$, so the result is $(4/9)(1/2) = 2/9$, independent of P.

例题解析

1. 若 M_i 为边中点，则对应三角形重心为 $T_i = (P + 2M_i)/3$。

2. 所以重心四边形是中点四边形按 $2/3$ 缩放的位似像。

3. 中点四边形面积为 $1/2$，故目标面积为 $2/9$，与 P 无关。

Writing one common center is stronger than checking separate side parallels.

写出共同位似中心，比逐一验证边平行更有力。

With circumcenter as origin, $h = a + b + c$ is the orthocenter vector. Reflection of the circumcircle across BC has center $b + c = h - a$.

以外心为原点时，垂心向量为 $h = a + b + c$，外接圆关于 BC 反射后的圆心为 $b + c = h - a$。

Practice / 配套练习: AIME-2003-II-Q04 (WB p. 438), AIME-2005-II-Q10 (WB p. 446), AIME-2006-I-Q10 (WB p. 369), AIME-2010-I-Q08 (WB p. 167), AIME-2018-II-Q09 (WB p. 379), AIME-2019-II-Q01 (WB p. 359), AIME-2020-I-Q15 (WB p. 408), AIME-2022-I-Q03 (WB p. 312), AIME-2026-II-Q12 (WB p. 435)

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Method 7 / 方法 7 [GE07-M07]

Half-plane clipping describes overlaps and nearest-point regions

半平面截割描述重叠及最近点区域

Procedure

1. Convert each boundary into a signed linear inequality; compare squared distances to turn nearest-point conditions into perpendicular-bisector half-planes.

2. Intersect every inequality and find feasible pairwise boundary intersections.

3. For a slope inequality, split by the sign of its horizontal denominator before multiplying.

4. Use the resulting actual polygon for area, not the unbounded supporting lines.

通用步骤

1. 将边界转为有向线性不等式；相减距离平方可把最近点条件转为垂直平分线半平面。

2. 相交全部不等式，求可行的两两边界交点。

3. 斜率不等式先按横向分母符号分区，再相乘。

4. 用实际截割多边形求面积，不能使用无界支撑直线。

> Example. Inside the unit square, find the area closer to $(0,0)$ than to either $(1,0)$ or $(0,1)$.

> 例题：单位正方形内，求比 $(1,0),(0,1)$ 都更靠近原点的区域面积。

Worked solution

1. Comparison with $(1,0)$ gives $x^2+y^2<(x-1)^2+y^2$, or $x<1/2$.

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坐标、向量与变换

Method 7 / 方法 7 [GE07-M07] (continued / 续)

2. Comparison with $(0,1)$ gives $y < 1/2$.

3. The intersection is a half-by-half square, area $1/4$. Strict boundaries have area zero.

例题解析

1. 与 $(1,0)$ 比较距离平方，得 $x < 1/2$。

2. 与 $(0,1)$ 比较，得 $y < 1/2$。

3. 交集为二分之一乘二分之一的正方形，面积 $1/4$；严格边界面积为零。

A factorized inequality can define several sign regions; identify the bounded one before measuring.

因式分解不等式可能给出多个符号区域，应先辨认有界部分。

For a constant-speed road followed by slower off-road travel, moving-center endpoint disks form a reachable envelope. First prove an optimal route can be reduced to the permitted road/straight-segment form, then derive the common tangent boundaries.

For a bounded lattice-cell union enclosed by a closed disk, pass to cell closures and check all polygon corners; convexity of the disk then covers each whole cell.

先沿高速道路、再慢速离路行驶时，移动圆心的终点圆盘组成可达包络。先证明最优路径可简化为允许的道路段加直线段，再求公切线边界。

闭圆盘包围有限格单元并集时，可先取各格闭包再检查全部顶点；圆盘凸性保证整格被包含。

Section 2.7 Coordinates, vectors & transformations

坐标、向量与变换

Method 7 / 方法 7 [GE07-M07] (continued / 续)

Practice / 配套练习: AIME-2000-I-Q13 (WB p. 384), AIME-2004-I-Q10 (WB p. 769), AIME-2010-I-Q11 (WB p. 458), AIME-2012-I-Q08 (WB p. 447), AIME-2013-I-Q12 (WB p. 398), AIME-2017-II-Q03 (WB p. 767), AIME-2020-II-Q02 (WB p. 768), AIME-2021-I-Q02 (WB p. 361), AIME-2025-I-Q12 (WB p. 413)

Section 2.7 Coordinates, vectors & transformations

坐标、向量与变换

Method 8 / 方法 8 [GE07-M08]

Supporting widths enforce full containment

支撑宽度施加整体容纳限制

Procedure

1. For a unit normal n, compute extreme projections $\min n \cdot v, \max n \cdot v$ of the entire figure.

2. Their difference is its width in that direction, giving strip or square-fit bounds.

3. For an inscribed disk, require the center's signed distance from every supporting side to be at least its radius.

4. After a formal optimum is found, check every side-segment condition and construct a feasible equality center.

通用步骤

1. 给定单位法向 n，求整个图形投影的最小及最大值。

2. 两者差为该方向宽度，给出条带或正方形容纳界。

3. 对内接圆，要求圆心到每条支撑边的向内有向距离至少为半径。

4. 求形式最优后，检查全部边上线段限制并构造可行取等圆心。

> Example. A unit square is rotated 45° about its center. Find the side of the largest concentric axis-parallel square inside it.

> 例题：单位正方形绕中心转四十五度，求其内部同心、边平行于坐标轴的最大正方形边长。

Section 2.7 Coordinates, vectors & transformations

坐标、向量与变换

Method 8 / 方法 8 [GE07-M08] (continued / 续)

Worked solution

1. If the inner side is t, its projection width onto a rotated-side normal is $t(\cos 45^\circ + \sin 45^\circ) = t\sqrt{2}$.

2. This must be at most the outer square's width one.

3. Thus $t \le 1/\sqrt{2}$, and the centered square of that side touches all four outer sides, attaining equality.

例题解析

1. 内方边长为 t 时，在外方边法向的投影宽为 $t\sqrt{2}$。

2. 它不能超过外方该方向宽一。

3. 故 $t \le 1/\sqrt{2}$，同心取此边长可同时接触四边，达到等号。

A line-intersection construction can lie beyond a side even when its equations are correct.

直线求交方程即使正确，构造点仍可能落在边的延长线上。

A side-t square rotated by θ has axis widths $t(|\cos\theta| + |\sin\theta|)$. For two concentric nested squares, apply the same support factor at both successive contacts.

For a symmetric semicircle, width in a nonparallel direction depends on both a straight-diameter endpoint and the curved support point. Derive the support function rather than treating the semicircle as a full circle.

边长 t 的正方形转角 θ 后，两坐标方向宽均为 $t(|\cos\theta| + |\sin\theta|)$；同心嵌套方形的相邻接触可连续应用同一支撑系数。

Section 2.7 Coordinates, vectors & transformations

坐标、向量与变换

Method 8 / 方法 8 [GE07-M08] (continued / 续)

半圆在非平行方向的宽度同时取决于直径端点与曲线支撑点，应推导支撑函数，不能把半圆当整圆。

Practice / 配套练习: AIME-2005-I-Q11 (WB p. 185), AIME-2005-I-Q14 (WB p. 391), AIME-2010-I-Q08 (WB p. 167), AIME-2013-I-Q12 (WB p. 398), AIME-2016-I-Q09 (WB p. 183), AIME-2016-II-Q07 (WB p. 376), AIME-2017-I-Q15 (WB p. 404), AIME-2018-I-Q08 (WB p. 378)

Section 2.7 Coordinates, vectors & transformations

坐标、向量与变换

Method 9 / 方法 9 [GE07-M09]

Primitive lattice directions and unfolded reflection rays

本原格点方向与反射展开射线

Procedure

1. Divide integer direction components by their GCD to obtain a primitive direction.

2. Unfold specular reflections so the entire path becomes one straight ray in repeated boxes.

3. A vertex return means every unfolded coordinate is a box-side multiple; solve the simultaneous divisibility conditions.

4. For a thickened lattice object, test its full linear-form range and correct shared endpoints when repeating a primitive segment.

通用步骤

1. 整数方向分量除以最大公因数，得到本原方向。

2. 展开镜面反射，使整个路径成为重复长方体中的一条直射线。

3. 回到顶点等价于展开各坐标均为边长倍数，应联立整除条件。

4. 厚格点物体应检查整个线性型范围，重复本原线段时还须修正共端点。

> Example. A ray in the cube $[0,2]^3$ starts at the origin in direction $(2,1,1)$ and reflects at every face. Find the distance to its first later vertex.

> 例题：光线在立方体 $[0,2]^3$ 中从原点沿 $(2,1,1)$ 方向发出并在各面反射，求首次再到顶点的路程。

Section 2.7 Coordinates, vectors & transformations

坐标、向量与变换

Method 9 / 方法 9 [GE07-M09] (continued / 续)

Worked solution

1. The unfolded ray is $(2t, t, t)$ with $t > 0$. A folded vertex requires all three coordinates divisible by two.

2. The first coordinate requires integral t, and the other two then require even t. The first is $t = 2$.

3. Distance is $2\sqrt{2^2 + 1^2 + 1^2} = 2\sqrt{6}$.

例题解析

1. 展开射线为 $(2t, t, t)$，折回顶点要求三坐标都为二的倍数。

2. 第一坐标要求 t 为整数，另两坐标进一步要求偶数，最小为二。

3. 路程为 $2\sqrt{6}$。

For radius-ε circles, use Euclidean normal length; for axis-aligned squares use the sum of absolute normal components.

半径 ε 圆使用法向量欧氏长度，轴平行正方形使用法向分量绝对值和。

A line with primitive direction v has successive lattice points separated by v. In the grid box $\{0, \ldots, N-1\}^d$, where N is a positive integer, containing k grid points requires $(k-1)|v_i| \le N - 1$ for every component. Normalize one endpoint to avoid counting a line repeatedly.

For square centers (u, v) near line $ax + by = 0$, a side-2ε square intersects the line when $|au + bv| \le \varepsilon(|a| + |b|)$. A radius-$\varepsilon$ circle instead uses $\varepsilon\sqrt{a^2 + b^2}$. Check segment endpoints separately.

Section 2.7 Coordinates, vectors & transformations

坐标、向量与变换

Method 9 / 方法 9 [GE07-M09] (continued / 续)

本原方向为 v 的直线，相继格点间差为 v。在格点盒 $\{0,\dots,N-1\}^d$ 中（N 为正整数），若直线含 k 个格点，则各分量满足 $(k-1)|v_i| \le N-1$；规范一个端点可避免重复计线。

格点中心 (u,v) 附近的边长 2ε 轴平行方形与 $ax+by=0$ 相交，当 $|au+bv| \le \varepsilon(|a|+|b|)$；半径 ε 圆则使用 $\varepsilon\sqrt{a^2+b^2}$，线段端点还须单独检查。

Practice / 配套练习: AIME-2000-II-Q11 (WB p. 385), AIME-2002-I-Q11 (WB p. 386), AIME-2016-I-Q14 (WB p. 403), AIME-2017-II-Q14 (WB p. 704), AIME-2020-II-Q12 (WB p. 409)

Section 2.7 Coordinates, vectors & transformations

坐标、向量与变换

Method 10 / 方法 10 [GE07-M10]

Complex inversion maps half-planes to disks

复数倒数将半平面映为圆盘

Procedure

1. Put $w = 1/z = u + iv$ and express z through u, v.

2. Translate the original half-plane inequality and multiply only by the positive quantity $u^2 + v^2$.

3. Complete squares to identify the image disk; preserve unions, intersections and excluded zero points.

通用步骤

1. 令 $w = 1/z = u + iv$，用 u, v 表示 z。

2. 翻译原半平面不等式，只乘正数 $u^2 + v^2$。

3. 配方识别像圆盘，保留并集、交集及零点排除条件。

> Example. Find the image of $\Re z > 1$ under $w = 1/z$.

> 例题：求半平面 $\Re z > 1$ 在 $w = 1/z$ 下的像。

Worked solution

1. Since $z = (u - iv)/(u^2 + v^2)$, the condition is $u/(u^2 + v^2) > 1$.

2. Thus $u^2 + v^2 - u < 0$, or $(u - 1/2)^2 + v^2 < 1/4$.

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Method 10 / 方法 10 [GE07-M10] (continued / 续)

3. The image is the open disk centered at $1/2$ of radius $1/2$; zero is on its boundary and is automatically excluded.

例题解析

1. 由 $z = (u - iv)/(u^2 + v^2)$，条件为 $u/(u^2 + v^2) > 1$。

2. 所以 $u^2 + v^2 - u < 0$，即 $(u - 1/2)^2 + v^2 < 1/4$。

3. 像为圆心 $1/2$、半径 $1/2$ 的开圆盘，零在边界上而自动被排除。

The exterior of a convex polygon is a union of exterior half-planes, so its inversion is a union of the corresponding disks.

凸多边形外部是各外侧半平面的并，故其倒数像为相应圆盘的并。

Practice / 配套练习: AIME-2008-II-Q13 (WB p. 90)

Section 2.7 Coordinates, vectors & transformations

坐标、向量与变换

Method 11 / 方法 11 [GE07-M11]

Develop a cylindrical path and preserve tangent pitch

展开圆柱路径并保持相切坡度

Procedure

1. Unroll the cylindrical surface into a plane with horizontal coordinate equal to circumferential arc length.

2. The shortest surface path is a straight segment in an appropriate winding lift.

3. For a taut rope leaving tangentially, extend that line's slope into its free portion and use the same length-to-horizontal-projection ratio.

通用步骤

1. 展开圆柱面，水平坐标取圆周弧长。

2. 适当绕行提升中的直线段给出最短表面路程。

3. 绷紧绳索相切离开时，其自由段延续展开直线坡度，并使用相同实际长度与水平投影比。

> Example. On a radius-one cylinder, two points are on opposite generators with height difference two. Find their shortest surface distance.

> 例题：半径一的圆柱上，两点位于相对母线、高度相差二，求最短表面距离。

Worked solution

1. The unfolded horizontal separation is $\pi + 2k\pi$ for integer winding number k.

2. The smallest absolute separation is π, attained for the two opposite winding choices.

Section 2.7 Coordinates, vectors & transformations

坐标、向量与变换

Method 11 / 方法 11 [GE07-M11] (continued / 续)

3. Pythagoras gives distance $\sqrt{\pi^2+4}$.

例题解析

1. 展开水平距离为 $\pi+2k\pi$，其中 k 为整数绕数。

2. 最小绝对水平距离为 π，对应两种相反绕向。

3. 由勾股，最短距离为 $\sqrt{\pi^2+4}$。

The free tangent length in horizontal projection is $\sqrt{d^2-R^2}$ for endpoint axis distance d.

端点距轴 d 时，自由切线水平投影长为 $\sqrt{d^2-R^2}$。

Practice / 配套练习: AIME-2004-I-Q14 (WB p. 454)

Section 2.7 Coordinates, vectors & transformations

坐标、向量与变换

Method 12 / 方法 12 [GE07-M12]

Orthonormal spatial coordinates separate height, plane and solid

标准正交空间坐标分离高度、平面及立体

Procedure

1. Choose three perpendicular solid-edge directions as a basis; encode the vertical direction by a unit vector.

2. Height differences along equal orthogonal edges have squared sum equal to the edge length squared.

3. Write cutting faces as plane equations and compute actual segment intersections, not just line intersections.

4. Use point–plane distances, determinants or projected-area factors according to the requested quantity.

通用步骤

1. 以三条垂直立体棱方向为基，用单位向量表示竖直方向。

2. 等长正交棱的高度差平方和等于棱长平方。

3. 把截面写为平面方程，并求实际棱线段交点，不仅是直线交点。

4. 按目标选择点面距离、行列式或投影面积系数。

> Example. A cube has a lowest vertex at height zero. Its three neighboring vertices have heights two,three,six. Find its volume.

> 例题：立方体最低顶点高度零，三个相邻顶点高度分别二、三、六，求体积。

Section 2.7 Coordinates, vectors & transformations

坐标、向量与变换

Method 12 / 方法 12 [GE07-M12] (continued / 续)

Worked solution

1. Let the edge length be s. The three vertical direction components are $2/s, 3/s, 6/s$.

2. Their squared sum is one, so $s^2 = 4 + 9 + 36 = 49$ and $s = 7$.

3. The volume is 343. All three components are positive, so the nominated vertex is indeed lowest.

例题解析

1. 设棱长 s，竖直单位向量三个分量为 $2/s, 3/s, 6/s$。

2. 平方和为一，所以 $s^2 = 49, s = 7$。

3. 体积为 343；三分量皆正，给定顶点确为最低点。

Plane incidence and a filled face's nearest point are different constraints.

平面关联与到填满面的最近点，是不同约束。

Common same-side tangent planes to spheres resting on $z = 0$ give $ux + vy - c = (1 - h)\rho$ for a unit normal (u, v, h). Excluding the repeated plane $h = 1$ makes radius an affine function of contact position; interpolate it from three noncollinear contacts and verify unit-normal feasibility.

球与 $z = 0$ 相切、又与同侧第二平面相切时，单位法向量 (u, v, h) 给出 $ux + vy - c = (1 - h)\rho$。排除重复平面 $h = 1$ 后，半径成为接触位置的仿射函数，可由三个不共线接触点插值，并需检验单位法向可行性。

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Method 12 / 方法 12 [GE07-M12] (continued / 续)

Practice / 配套练习: AIME-2001-II-Q15 (WB p. 451), AIME-2002-I-Q15 (WB p. 452), AIME-2006-I-Q14 (WB p. 393), AIME-2007-I-Q13 (WB p. 456), AIME-2008-I-Q15 (WB p. 457), AIME-2011-I-Q13 (WB p. 432), AIME-2012-I-Q08 (WB p. 447), AIME-2013-I-Q07 (WB p. 373), AIME-2017-II-Q15 (WB p. 405), AIME-2023-I-Q13 (WB p. 460), AIME-2023-II-Q14 (WB p. 461), AIME-2026-I-Q12 (WB p. 434), AIME-2026-II-Q12 (WB p. 435)

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坐标、向量与变换

Method 13 / 方法 13 [GE07-M13]

Coordinate area uses the boundary's true order

坐标面积使用真实边界次序

Procedure

1. Compute intersection points exactly and arrange them in actual boundary order.

2. Use a planar determinant or shoelace sum for a simple polygon, and a cross-product norm for a spatial triangle.

3. For a crossed outline, partition all bounded lobes before adding positive areas.

4. Normalize by an affine map only when the common determinant factor cancels from an area ratio.

通用步骤

1. 精确求交点，并按实际边界次序排列。

2. 简单多边形用行列式或鞋带和，空间三角形用叉积长度。

3. 交叉轮廓需分割全部有界叶片，再加正面积。

4. 只有共同面积行列式因子可约去时，才作仿射规范。

> Example. Find the area of triangle $(1,1),(5,2),(2,4)$.

> 例题：求三角形 $(1,1),(5,2),(2,4)$ 的面积。

Worked solution

1. From the first vertex, the two side vectors are $(4,1),(1,3)$.

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坐标、向量与变换

Method 13 / 方法 13 [GE07-M13] (continued / 续)

2. Their determinant is $4 \cdot 3 - 1 \cdot 1 = 11$.

3. The triangle area is $|11|/2 = 11/2$.

例题解析

1. 从第一顶点出发，两边向量为 $(4, 1), (1, 3)$。

2. 行列式为 $4 \cdot 3 - 1 \cdot 1 = 11$。

3. 三角形面积为 $11/2$。

A global absolute shoelace value is an algebraic area for a self-intersecting polygon.

自交多边形的全局鞋带绝对值仍是代数面积。

Practice / 配套练习: AIME-2005-I-Q10 (WB p. 368), AIME-2010-II-Q09 (WB p. 372), AIME-2013-I-Q07 (WB p. 373), AIME-2013-II-Q13 (WB p. 399), AIME-2014-I-Q13 (WB p. 400), AIME-2015-I-Q04 (WB p. 357), AIME-2018-II-Q04 (WB p. 358), AIME-2018-II-Q09 (WB p. 379), AIME-2019-II-Q01 (WB p. 359), AIME-2025-I-Q02 (WB p. 245), AIME-2025-I-Q12 (WB p. 413), AIME-2026-I-Q10 (WB p. 383), AIME-2026-II-Q03 (WB p. 365)

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Method 14 / 方法 14 [GE07-M14]

Velocity vectors distinguish the moving and ground frames

速度向量区分运动介质与地面参照系

Procedure

1. Write ground velocity as medium-relative velocity plus the medium's velocity.

2. Use the target displacement divided by time for the ground-frame components.

3. Apply speed norms in the correct frame, and subtract squared equations to eliminate the shared transverse component.

通用步骤

1. 地面速度等于相对介质速度加介质速度。

2. 由目标位移除以时间写出地面分量。

3. 在正确参照系下使用速率范数，并相减平方式以消去共同横向分量。

> Example. A river is twelve units wide and flows east at one unit per minute. A swimmer of water-relative speed $\sqrt{5}$ travels directly north in the ground frame. Find the crossing time.

> 例题：河宽十二，水流向东速度一单位每分，泳者相对水速为 $\sqrt{5}$，地面轨迹正北，求过河时间。

Worked solution

1. To cancel east current, the swimmer's water-relative horizontal component must be -1.

2. If its north component is $v > 0$, then $1 + v^2 = 5$, so $v = 2$.

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Method 14 / 方法 14 [GE07-M14] (continued / 续)

3. The ground north speed remains two, giving time $12/2 = 6$ minutes.

例题解析

1. 抵消向东水流要求相对水的横向分量为 -1。

2. 若向北分量为 $v > 0$，则 $1 + v^2 = 5$，所以 $v = 2$。

3. 地面向北速度仍为二，故过河时间为六分钟。

The magnitude of a sum of velocities is not generally the sum of their magnitudes.

速度向量和的长度一般不等于速率之和。

Practice / 配套练习: AIME-2022-I-Q05 (WB p. 20)

Section 2.7 Coordinates, vectors & transformations

坐标、向量与变换

Method 15 / 方法 15 [GE07-M15]

Oblique unit-vector bases retain a Gram metric

斜交单位向量基保留格拉姆度量

Procedure

1. Choose linearly independent unit side directions u, v and record $u \cdot v = c$, with $|c| < 1$.

2. A point $xu + yv$ has norm square $x^2 + y^2 + 2cxy$; incidence stays affine but Euclidean lengths use this quadratic form.

3. The vector $u + v$ points along the internal bisector because its dot products with the two unit sides are equal.

4. Subtract circle equations sharing this metric form to obtain a linear radical axis.

通用步骤

1. 取线性无关的单位边方向 u, v，记录 $u \cdot v = c$，其中 $|c| < 1$。

2. 点 $xu + yv$ 的范数平方为 $x^2 + y^2 + 2cxy$；关联保持仿射，但欧氏长度必须使用此二次型。

3. $u + v$ 对两单位边的点积相同，因此指向内角平分线。

4. 同一度量下的两圆方程相减，仍得到线性根轴。

> Example. Unit vectors u, v meet at 60°. Find $|2u + v|$ and a unit vector along their internal bisector.

> 例题：单位向量 u, v 夹角六十度，求 $|2u + v|$ 及其内角平分线单位方向。

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坐标、向量与变换

Method 15 / 方法 15 [GE07-M15] (continued / 续)

Worked solution

1. The Gram coefficient is $u \cdot v = 1/2$. Thus $|2u+v|^2 = 4+1+4(1/2) = 7$.

2. Also $|u+v|^2 = 2+2(1/2) = 3$, so its unit direction is $(u+v)/\sqrt{3}$.

3. The direction has equal dot product $\sqrt{3}/2$ with u, v, confirming that each angle is 30°.

例题解析

1. 格拉姆系数为 $1/2$，所以 $|2u+v|^2 = 4+1+4(1/2) = 7$。

2. 又 $|u+v|^2 = 3$，故单位方向为 $(u+v)/\sqrt{3}$。

3. 该方向与 u, v 点积均为 $\sqrt{3}/2$，核实两角各三十度。

Using x^2+y^2 in an oblique basis silently changes the geometry.

斜交坐标中若使用 x^2+y^2，便悄悄改变了几何度量。

For a midpoint M of side AB, with $C = 0, A = bu, B = av$, subtracting $\min(a,b)(u+v)/2$ from M locates the opposite perimeter-bisector endpoint on the longer adjacent side. The two boundary routes then have equal length; this proves the cleaver direction, rather than assuming it.

若 M 为 AB 中点，取 $C = 0, A = bu, B = av$，从 M 减去 $\min(a,b)(u+v)/2$，便得到较长邻边上的周长平分线另一端点。两条边界路线长度相等，从而证明劈周长线方向，而非直接假设该结论。

Practice / 配套练习: AIME-2010-II-Q15 (WB p. 296), AIME-2022-I-Q14 (WB p. 326)